Everyone here knows about the replication crisis. Only 36% of papers in psychology turned out to be reproducible and it shook up the social sciences. Which led us to think about the question: if reproducibility is so low, how can we trust science?

Well, my name is Anna Vijlbrief and today I will talk to you about my thesis which addresses one solution to the replication crisis: The preregistration.

I will tell you about why we have preregistration, what the problem with preregistrations is, what we don't know and how my research fits into this.

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During research, a lot of decisions need to be made, such as 'how many participants will I collect' and 'how will I deal with outliers'. Researchers have a lot of freedom to decide how they wish to deal with these questions and these decisions are often made during the research process. However, we want these decisions to be independent of the data. When researchers, knowingly or unknowingly, start making decisions based on what gives them the best results, this can lead to not reproducible findings.
Preregistrations were proposed as a solution to prevent this. You record your research choices in a published document beforehand and then stick to your premade plan during the research.

The problem with preregistrations however, is that people deviate from them. They record their research process beforehand but then end up changing it during their study. Deviations occur in as much as 93% of cases but even worse, 63% of those deviations go unreported. This means that only in very few cases are people transparent about the fact that they did not stick to their preregistered plans. For example if all of us preregistered our theses, only Paolo would stick to his preregistered plan. The rest of us would all deviate from our plans, and most of us would not be transparent about it.

As a result of this, preregistration might not limit researcher degrees of freedom at all. And researchers might still be searching for significant results, even when they preregister.

So we know that people preregister, and we know that people deviate, but what we don't know what **the effect** of these deviations is? Is it good to deviate by collecting more participants than you originally planned? Is it bad to include an unplanned covariate? How bad is it to deviate? People have theorized about these effects, both negatively and positively but they have yet to be examined empirically. And that is what I will be doing in my research. More specifically, I aim to investigate 'How do deviations from preregistrations affect type I and type II error rates'.

And I am doing that as follows. In my project I will generate data according to a simple linear regression model. Data will be generated in two scenarios, either a no effect scenario or an effect scenario. Within each scenario I will then simulate different types of deviations and compare the type 1 and 2 error rates to a baseline condition without any deviations.

In order to identify my conditions, I sought out important deviations in psychology. This led me to three important domains in which deviations commonly occur: sample size, outlier identification and the statistical model. Within these domains I identified the following deviations which will be the basis for my simulation conditions: within the sample size domain I will have small and big sample size deviations, for outliers I will look at z-scores and cook's distance and for the statistical model I will deviate by adding covariates and by switching outcomes.

What I expect my results to look like is that for each simulation condition, I will be able to assess the type I/II error rate inflations. In the sample size and outlier domains, these results are more directly comparable, for example like this:

The baseline scenario would be expected to have a nominal type I error rate of 5%, whereas the deviation conditions might have larger error rates.
In the statistical model condition I will probably compare the covariate conditions and the outcome switching condition separately.
I will also be able to provide an overview of which of these deviations has the largest impact on type I/type II error.
These are all mock results as I have not yet run my actual simulation. But

What I hope to offer with this simulation is a systematic research of deviations. As well as clarity on the effect of deviations from preregistration on type I and type II errors. Hopefully this will also help us further assess the impact of deviations. Because we know deviations are very common, but the question remains: are they actually bad?

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If time allows I would like to extend this project. The reason behind why people deviate can be quite important. Did someone choose to deviate because it led to 'better' results or did they have no other choice because for example a model assumption was violated? To include an element of choice in my research I plan to extend this project by running an additional simulation in which I let the simulate choose either the baseline outcome or the deviation outcome. I hope that by doing this I can assess the impact of choice on how big the effect of a deviation is.

, as well as the effect of picking deviations opportunistically

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